



Research articles

Multi-objective optimization of a reluctance actuator for precision motion applications

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ABSTRACT

Recent research and assessments have shown that the reluctance actuator is a promising candidate to drive the next generation precision motion system of the lithography machine in the semiconductor industry. This paper proposes a multi-objective optimization algorithm for an optimal design of E-core reluctance actuator considering the requirements of the precision motion system. The objectives of optimization include maximizing the output magnetic force, minimizing the mass and the time constant of the actuator, and driving the actuator stiffness value to zero. First, the characteristics of the reluctance actuator are expressed in terms of the design variables of the reluctance actuator. After that, the optimization algorithm using the Grid search method and Pareto optimal approach is proposed to obtain the optimal design of the reluctance actuator. Finally, the optimal design of the reluctance actuator is implemented in a precision motion system example. The simulation results are obtained using Finite Element Analysis in COMSOL to check the saturation of magnetic flux in the reluctance actuator. The dynamic response of reluctance actuator motion system is obtained in the time domain for different inputs to verify the optimal design of the reluctance actuator.

1. Introduction

Design and development of a precision motion system play an important role in the future micro and nanotechnologies in industrial applications, such as the lithography machine in semi-conductor industry [1]. The precision motion system consists mainly of a movable stage moved in a work-space within a micro-displacement range and sub-nanometers resolutions [2]. The semiconductor manufacturing industry requires an increasing throughput (wafer per hour) and accuracy (micro-meters). To achieve these requirements, the force should be enhanced (stable and accurate) to generate faster acceleration of short-stroke motion system in semiconductor lithography machines, see for example [2,3].

Electromagnetic actuators are widely used actuators to drive the high precision positioning motion systems in different applications, see for example [4–7]. Electromagnetic actuators are transducers that convert electrical energy into a movement within a limited linear or rotational range. The magnetic force of the electromagnetic actuators is categorized into Lorentz force where the magnetic force is linearly proportional to the current and Reluctance force where the force is proportional to the square of the current [2]. A reluctance actuator is an electromagnetic actuator that consists of ferromagnetic stationary and movable cores (stator and mover). The reluctance actuator generates

a uni-directional attractive magnetic force in the air gap between the stationary core (stator) and the movable core (mover), this force propels the mover toward the stator. The reluctance actuator can achieve higher force density with low actuator mass and low heat dissipation when compared with the Lorentz actuator. The main challenging task in the reluctance actuator is how to handle the nonlinear relationship between the magnetic force and the current, and between the magnetic force and the air gap, on the other hand [8]. These nonlinearities cause a serious problem in the design, control, and operation of the reluctance actuators, which limits using them in the precision motion system.

Recently, the reluctance actuators have been developed to operate positioning motion systems that required high force within limited workspace. These systems include reticle and wafer stages in lithography machines in semiconductor manufacturing industry [9,10], magnetic levitation systems, and maglev transportation systems [11]. It is important to mention that the literature focused on mathematical modeling to determine the relationship of the magnetic force of the reluctance actuator as a function of the input current and the air gap, see for example [9,10,12,13]. However, in real-time motion control applications, the reluctance actuator is connected to a motion system, where the mover is a part of the considered motion system. Thus, the magnetic force depends on the motion system dynamics and the air gap.

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Section 2 develops the reluctance actuator dynamic model considering the motion system dynamics.

Optimization techniques have an essential role in improving the performance of electromechanical devices, for example, see [14–16]. Different techniques and methods are presented in several studies for topology optimization of magnetic actuators. For example, non-dominated sorting genetic algorithm [17,18], adaptive weight determination scheme and tunneling method [19], gradient-based optimization algorithm and an adaptive weight determination scheme [20], and hybrid approach of the equivalent circuit method and the finite analysis method [21]. In these studies, topology optimization of the magnetic actuator was formulated as a multi-objective design problem, and the main objective of the optimization is to maximize the magnetic force and to minimize the volume of the actuator. In our proposed design optimization algorithm, the objective function maximizes the force and minimizes the mass taking into account the precision motion requirements including desired maximum force and specific range of motion. In this work, the optimization algorithm based on the direct grid search method and Pareto-optimal approach is used to determine the optimal design variables of the reluctance actuator. First, the direct optimization based on the grid search method is considered to obtain all suggested combinations of the design variables of the optimization problem. Then, the Pareto-optimal approach determines the Euclidean distance between all the suggested designs and the ideal point in the Pareto-optimal curve. Finally, the optimal design is selected with the minimum Euclidean distance from the Pareto-optimal curve.

The proposed algorithm showed a helpful solution for the designers to obtain the optimal design variables of a reluctance system considering multi-objective criteria, such that the desired force, minimum mass, and specific air gap range, which represent the high precision motion requirements. The magnetic simulation results using Finite Element Analysis (FEA) showed that the flux density distribution within the actuator core is limited by the saturation value, thus, the design respects the magnetic limitations, and allows the operation to be in the linear region. Selection of the optimal design variables of the reluctance actuator linearizes the behavior of the reluctance actuator, which will approximately have the permeability of the linear region of the BH curve and will reduce the all-over non-linearity in the system. Also, this makes the approximation of the magnetic force is valid, which allows designing and implementing a feedforward controller of the reluctance system. Selection of the optimal design of the reluctance actuator is an essential step to building an experimental prototype of the system. It helps to choose the best design that satisfies the required specifications without over-design or under-design possibilities. It makes modifications without the need to obtain more than one experimental prototype. The prototyping and experimental work would be the next stage after optimization and design variables.

Up to date, the reluctance actuator design variables method has not been standardized yet. However, to evaluate the effectiveness of the proposed design algorithm, the reluctance actuator with optimal design variables is implemented to drive a motion system with a displacement of 2 mm and required force of 400 N. The novelty of this work is to consider the dynamics of the motion system (attached to the reluctance actuator) within the reluctance actuator design. Specifically, providing the optimal design variables to operate the reluctance actuator in the linear region of the BH-curve and to satisfy the requirements of the high precision motion system. These requirements include the desired force, the minimum stiffness, the minimum actuator mass, and the desired workspace range. Such a design is essential for the next generation of semiconductor manufacturing machines. The main contributions are to (i) present the optimal design of the reluctance actuator as a multi-objective optimization problem, considering the high precision motion requirements. (ii) propose an optimization algorithm based on the direct grid search method to determine all suggested design variables, then determine the optimal design using the Pareto-optimal approach.

(iii) present a dynamic response of a precision motion system driven by the reluctance actuator.

This paper is organized as follows. The characteristics of the reluctance actuator are described in details in Section 2. The design of reluctance actuator is proposed as multi-objective design optimization problem in Section 3. This section also presents the proposed design optimization algorithm using the Grid search method and Pareto optimal approach. Section 4 shows a numerical example with desired design requirements to evaluate the usefulness of the proposed optimization algorithm. The simulation results of the dynamic behavior of the precision motion system driven by the reluctance actuator are presented in Section 5. Finally, the conclusion is presented in Section 6.

2. Description of the reluctance actuator

This section presents the description of the motion system attached to an E-core reluctance actuator. Then, the main design variables of the reluctance actuator are presented to characterize the output magnetic force. Also, this section presents the dynamic model of the motion system attached to the reluctance actuator.

2.1. Design variables and magnetic properties

There are two main topologies of reluctance actuators, C-Core and E-Core topologies. In this work, the E-Core topology is considered. A schematic diagram of an E-core reluctance actuator with design variables is shown in Fig. 1(a). The design variables are presented in Table 1, and expressed in terms of a number of turns in the coil N and a width ratio constant r_b .

This section uses the Magnetic Equivalent Circuit (MEC) to formulate the analytical expression of the magnetic flux and the magnetic force as functions the design variables [22,23]. The Ampere's Circuital Law for E-core reluctance actuator shown in Fig. 1 considering the eddy current field is given by

$$\oint H \cdot dl = Ni \quad (1)$$

$$H_f(l_{s1} + 2l_{s2} + l_m) + H_e(l_{s1} + 2l_{s2} + l_m) + 3gH_g = Ni \quad (2)$$

where $H_e = \frac{\sigma^2}{12} \frac{dB}{dt}$ is the magnetic field of the eddy current [24], σ is the material electrical conductivity, t is the lamination thickness, $H_f = \frac{B}{\mu_r(B,H)\mu_0 A_{aa}}$ is the magnetic field in the core, and $H_g = \frac{B}{\mu_0}$ is the magnetic field in the air gap. Substitute H_e , H_f and H_g in (2), the magnetic model of the reluctance actuator is given by

$$(l_{s1} + 2l_{s2} + l_m) \frac{B}{\mu_r(B,H)\mu_0 A_{aa}} + \frac{\sigma t^2 (l_{s1} + 2l_{s2} + l_m)}{12} \frac{dB}{dt} + \frac{3g}{\mu_0} B = Ni, \quad (3)$$

The magnetic cores (mover and stator) of the reluctance actuator are made of thin iron sheets coated (stacks) with an insulating layer. These layers are in parallel with the magnetic flux lines, which minimize the effect of the eddy current. Then, the magnetic circuit analysis of the reluctance actuator can be formulated by neglecting the effects of the eddy current and the associated magnetic force.

Using the magnetic circuit analysis for MEC in Fig. 1(b), the magnetic flux Φ can be expressed by

$$\Phi = \frac{Ni}{(\mathcal{R}_{st1} + 0.5\mathcal{R}_{st2} + 0.25\mathcal{R}_m) + (\mathcal{R}_{g1} + 0.5\mathcal{R}_{g2})} = \frac{Ni}{\mathcal{R}_{core} + \mathcal{R}_{gap}}, \quad (4)$$

where the reluctances are defined as $\mathcal{R}_{st1} = \frac{l_{st1}}{\mu_r(B,H)\mu_0 A_{aa}}$, $\mathcal{R}_{st2} = \frac{l_{st2}}{\mu_r(B,H)\mu_0 A_{aa}}$, $\mathcal{R}_m = \frac{l_m}{\mu_r(B,H)\mu_0 A_{aa}}$, $\mathcal{R}_{g1} = \frac{g}{\mu_r(B,H)\mu_0 A_{ab}}$, and $\mathcal{R}_{g2} = \frac{g}{\mu_r(B,H)\mu_0 A_{ab}}$.

Substitute the aforementioned reluctance expressions in (4), then the magnetic flux can be as

$$\Phi = \mu_0 A_{aa} N \frac{i}{\left(\frac{1}{r_b} l_{st1} + \frac{1}{2} l_{st2} + \frac{1}{4} l_m \right) \mu_r(B,H) + \left(\frac{1}{r_b} + \frac{1}{2} \right) g}, \quad (5)$$

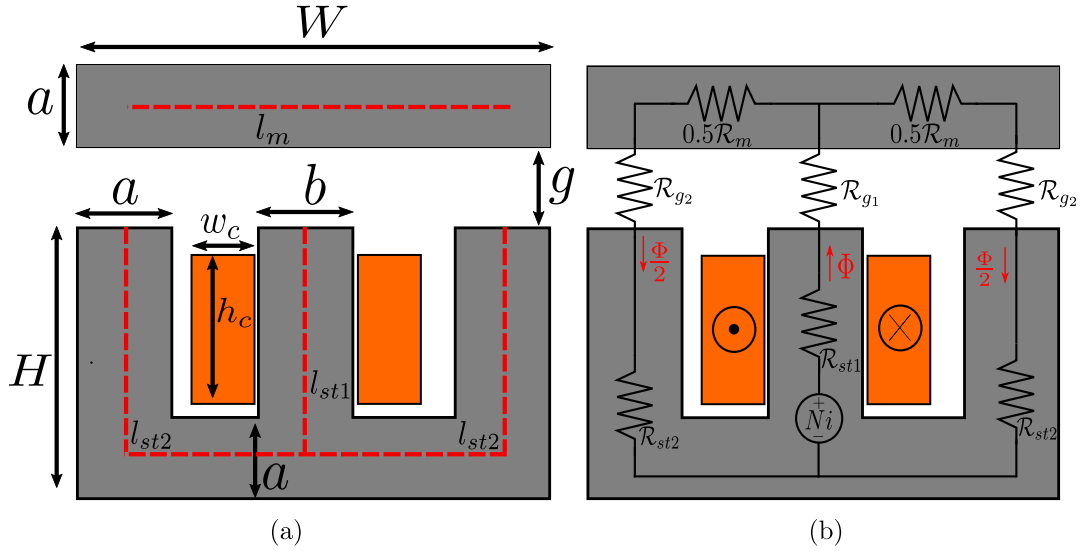


Fig. 1. (a) Schematic diagram of the E-core reluctance actuator with the design variables given in Table 1, and (b) magnetic equivalent circuit of the reluctance actuator in (a).

Table 1
Design variables of the reluctance actuator represented by schematic diagram in Fig. 1.

Symbol	Description	Value
w_c	Coil width	$d \times \text{ceil} \left(\frac{\sqrt{N/2}}{\sqrt{N/2}} \right) \times k_b$ k_b is build factor in range of [1.05, 1.4]
h_c	Coil height	$d \times \text{ceil} \left(\frac{N}{\sqrt{N/2}} \right) \times k_b$ k_b is build factor in range of [1.05, 1.4]
a	Two poles width	$w_c + c$
b	Center pole width	$r_b a$
W	Actuator width	$2a + b + 2(w_c + c)$
H	Actuator height	$a + (h_c + c)$
c	Clearance	1mm
g	Air gap	given range (desired workspace range)
N	Number of coil turns	optimized variable
r_b	Width ratio constant	$\frac{b}{a}$ optimized variable

where N is the number of turns in the coil, $A_{aa} = a \times a$ is the cross-section area, i is the input current, $r_b = \frac{b}{a}$ is the width ratio between center pole and side pole, g is the air gap between the I-core (mover) and the E-core (stator), $\mu_r(B, H)$ is the relative permeability of the ferromagnetic material defined by BH-curve, μ_0 is the permeability of the free space, l_{st1} and l_{st2} are the mean path lengths of the magnetic flux in the E-core, and l_m is the mean path lengths of the magnetic flux in the I-core as illustrated in Fig. 1(a). Based on Maxwell's stress tensor, the magnetic force in the air gap between the E-core and I-core is expressed by [23]

$$F = \frac{1}{2\mu_0} \left(\frac{1}{r_b} + \frac{1}{2} \right) \frac{\Phi^2}{A_{aa}}. \quad (6)$$

Substitute (5) in (6), the generated magnetic force can be expressed as

$$F = \frac{\mu_0 A_{aa} N^2}{2} \left(\frac{1}{r_b} + \frac{1}{2} \right) \frac{i^2}{\left[\frac{\frac{1}{r_b} l_{st1} + \frac{1}{2} l_{st2} + \frac{1}{4} l_m}{\mu_r(B, H)} + \left(\frac{1}{r_b} + \frac{1}{2} \right) g \right]^2}. \quad (7)$$

The above equation presents the relationship between the design variables and the output magnetic force.

2.2. Dynamic modeling of the reluctance actuator motion system

For different high precision applications, electromagnetic magnetic actuators have been proposed to drive a movable stage guided by a flexure hinge mechanism [25–27]. The schematic diagram of the motion

system driven by the reluctance actuator is shown in Fig. 2(a). This system consists of E-core reluctance actuators attached to a movable stage guided through a flexure hinge mechanism. In this system, the current i through the coil N generates an attractive magnetic force F in the air gap g between the E-core stator and I-core mover. This force moves the movable stage m toward the reluctance actuator in the +ve direction of X -axis ($x > 0$).

The mathematical model of the reluctance actuator with motion system is formulated to describe the electrical, magnetic, and mechanical properties of the system. The lumped parameter model of the electromechanical representation of the reluctance actuator motion system is illustrated in Fig. 2(b). In this model, the electrical parts are the supplied voltage $v(t)$ and the coil resistance Rm , and the mechanical parts are the movable stage mass m , the stiffness coefficient k , and the damping coefficient b . The energy conversion between the electrical domain and the mechanical domain takes place through the magnetic flux in the reluctance actuator cores. The electromechanical dynamic model of the reluctance actuator motion system is expressed by the ordinary differential equations as

$$N \frac{d\Phi(t)}{dt} + Ri(t) = v(t), \quad (8a)$$

$$\Phi(t) = \frac{N}{R_{core} + R_{gap}(g)} i(t), \quad (8b)$$

$$m \frac{d^2 x(t)}{dt^2} + b \frac{dx(t)}{dt} + k_h x(t) = F(t), \quad (8c)$$

$$F(t) = \frac{1}{2\mu_0 A_{aa}} \left(\frac{1}{r_b} + \frac{1}{2} \right) \Phi^2(t), \quad (8d)$$

$$g(t) = g_{max} - x(t). \quad (8e)$$

where $v(t)$ is the applied voltage, $i(t)$ is the current through the coil N , $\Phi(t)$ is the magnetic flux, $F(t)$ is the magnetic force in the air gap g , and $x(t)$ is the displacement of the movable mass m . The movable stage has mass m , stiffness coefficient k_h of the flexure hinge, and damping coefficient b . Fig. 3 shows the lumped parameter representation of dynamic model of the reluctance actuator motion system.

3. Multi-objective design optimization of reluctance actuator

In this section, the design of reluctance actuator is formulated as a multi-objective optimization problem. Mathematically, the multi-objective optimization problem can be formulated as $\min \{f_1(s), f_2(s), f_3(s), \dots, f_Q(s)\}$ subject to : $s \in S$, where $f_Q(s)$ is the objective function with $Q \geq 2$, and $s = (p_1, p_2, \dots, p_m)$ is the design variables vector

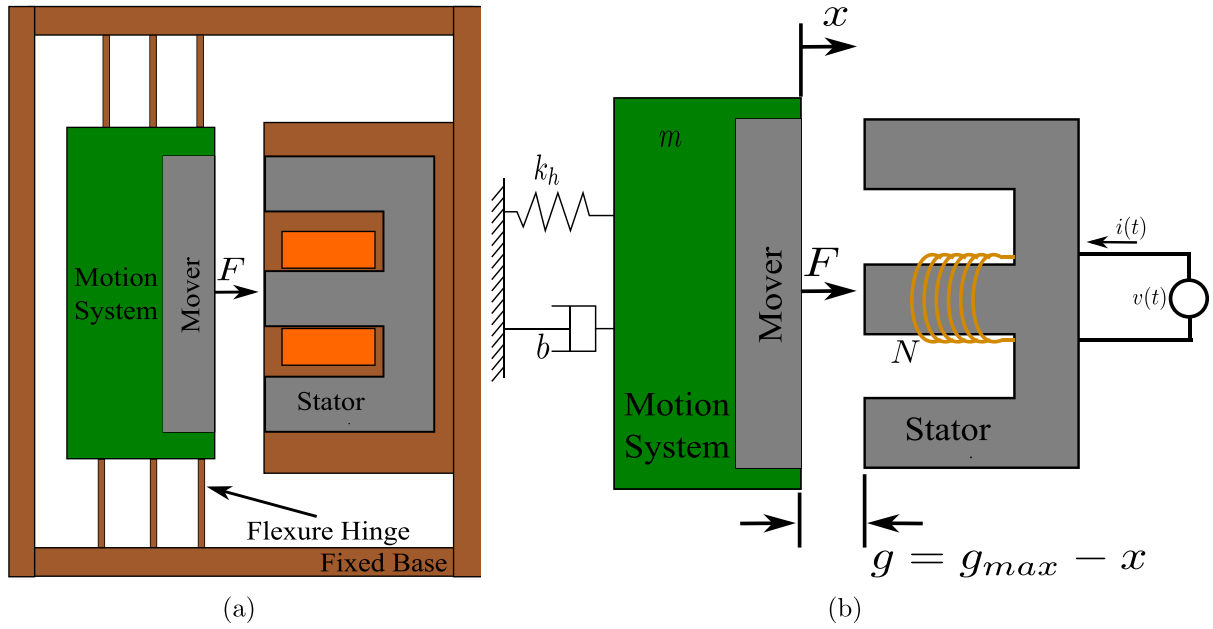


Fig. 2. (a) The schematic representation of the motion system attached to the reluctance actuator, and (b) lumped model of the motion system attached to the reluctance actuator.

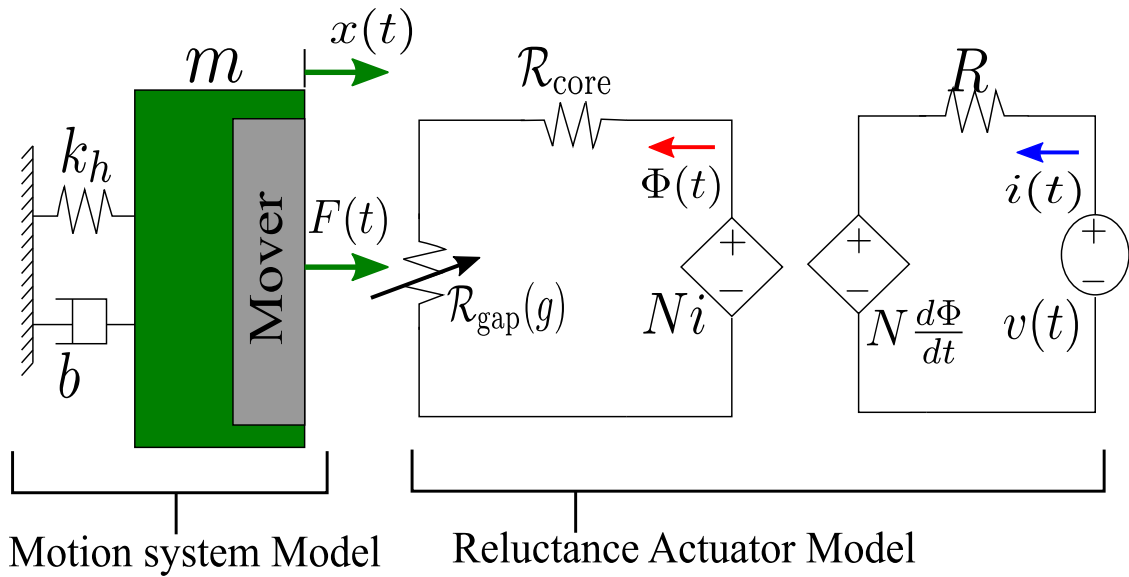


Fig. 3. Lumped parameters representation of the dynamic model of the reluctance actuator motion system.

in search space S . This section presents the formulation the objective functions related to the design optimization of the reluctance actuator. Then, the proposed optimization algorithm using Grid search method and Pareto optimal approach is characterized.

3.1. Optimization objective functions

The objective functions for the proposed optimization algorithm are selected to satisfy the precision motion system requirements. These requirements, including desired force, minimum actuation stiffness, minimum mass, and minimum time constant, have been obtained based on mathematical modeling. The objective functions are subjected to the following constraints: (i) generating the desired force over a range of air gaps, and (ii) the maximum input current is limited by the allowable current of the coil wire gauge. The mathematical formulation of a

multi-objective optimization to determine the optimal design variables $s = \{N, r_b\}$ of the reluctance actuator is given by:

$$\begin{aligned} & \max \{f_{k_E}(s)\}, \min \{f_M(s), f_L(s), f_R(s)\} \\ & \text{Subject to search space: } s \in S; \\ & \text{Subject to constraints: } F = F_d; i \leq i_{mc} \end{aligned} \quad (9)$$

where $f_{k_E}(s)$ is the objective function related to the actuator stiffness, $f_M(s)$ is the objective function related to actuator mass, $f_L(s)$ is the objective function related to the equivalent inductance, $f_R(s)$ is the objective function related to coil resistance, F is the magnetic force in (7), F_d is the desired force, i is the input current, i_{mc} is the maximum allowable current of coil gauge, $s = \{N, r_b\}$ is the design variables vector, S is the search space. The objective functions are formulated based on the reluctance actuator characteristics as follows.

3.1.1. Actuator stiffness

The stiffness represents the variation of magnetic force generated between the E-core and the I-core to the variation of the air gap distance [28]. The stiffness is expressed as $k_E = \frac{\Delta F}{\Delta g} = \frac{F - F_0}{g - g_0}$, where F_0 is the magnetic force at air gap g_0 and F is the magnetic force at air gap g . Considering small variation in the air gap $\Delta g \approx 0$, then the air gap $g \approx g_0$. Using the magnetic force in (7), the stiffness $k_E = \frac{\partial F}{\partial g}$. Then, the objective function of the stiffness can be expressed as

$$f_{k_E} = k_E = -\mu_0 A_{aa} N^2 \left(\frac{1}{r_b} + \frac{1}{2} \right)^2 \frac{i^2}{\left[\left(\frac{\frac{1}{r_b} l_{s1} + \frac{1}{2} l_{s2} + \frac{1}{4} l_m}{\mu_r(B, H)} \right) + \left(\frac{1}{r_b} + \frac{1}{2} \right) g \right]^3}, \quad (10)$$

It can be noticed that the stiffness f_{k_E} is directly proportional with the square of the input current g and inversely proportional with the air gap g . The negative sign of the stiffness f_{k_E} means that the magnetic force decreases as the air gap increases. Thus, the reluctance actuator stiffness causes a decrease in the overall stiffness of the motion system. This decrease contributes to unstable dynamic behavior if the magnetic stiffness exceed the stiffness provided by the flexure hinge mechanism. Thus, it is essential to maximize the stiffness of the reluctance actuator and drive it to the zero.

3.1.2. Actuator mass

The total mass of the reluctance actuator consists of the E-core stator mass M_{stator} , the I-core mover mass M_{mover} , and the coil mass M_{coil} as

$$f_M = M = M_{stator} + M_{mover} + M_{coil}. \quad (11)$$

From basic geometry and dimension of the reluctance actuator, the E-core stator mass is expressed by $M_{stator} = \rho_{cr} [a((H \times W) - 2(h_c + c) \times (w_c + c))]$, and the I-core mover mass is $M_{mover} = \rho_{cr} [a(a \times W)]$, where ρ_{cr} the density of the ferromagnetic material. The coil mass M_{coil} depends on the shape and dimension of the used coil as $M_{coil} = k_{pf} \rho_{cl} [h_c (\pi(w_c)^2 + (2b + 2a)w_c)]$, where ρ_{cl} is the density of coil wire material, and $k_{pf} = \frac{N\pi(\frac{d}{2})^2}{h_c w_c}$ is the packing factor, where d is the diameters of the coil wire [22]. The reluctance actuator is proposed to drive the precision motion system. In this design, the I-core mover of the reluctance actuator is part of the stage. Thus, the design of the reluctance actuator with a minimum size of I-core mover will increase the acceleration of the stage. Then, it is essential to minimize the actuator mass.

3.1.3. Coil equivalent inductance and resistance,

In magnetic actuator, the inductance depends on the mover position and this effects on the dynamic behavior of the actuator [29]. In magnetic circuit, the equivalent inductance is defined by $L = N \frac{\partial \Phi}{\partial i}$ [30]. Then, the equivalent inductance can be expressed

$$f_L = L = \frac{\mu_0 A_{aa} N^2}{\left(\frac{\frac{1}{r_b} l_{s1} + \frac{1}{2} l_{s2} + \frac{1}{4} l_m}{\mu_r(B, H)} \right) + \left(\frac{1}{r_b} + \frac{1}{2} \right) g} \quad (12)$$

The coil resistance is expressed as function of the coil parameters [22]

$$f_R = R = \frac{l}{\sigma A_{aa}} = \frac{V_{cl} N^2}{\sigma k_{pf} h_c^2 w_c^2}, \quad (13)$$

where σ is the coil conductivity. The reluctance actuator can be modeled as an RL circuit with a time constant $\tau = L/R$. This time indicates the required time to reach the steady-state value of the required force [31]. The precision motion system has a setting time in milliseconds [32]. Hence, the time constant of the reluctance actuator must be less than the settling time of the precision motion system. The time constant is proportional to the inductance L and inversely

proportional to the resistance R . Therefore, the optimized design of the reluctance actuator requires to decrease the inductance L and increase the resistance R . Increasing the resistance R contributes to increasing the copper losses of the coil. Then, the resistance R should be optimized to obtain the minimum possible time constant and copper losses.

3.2. Optimization algorithm

The proposed optimization algorithm combines the grid search method with Pareto optimal approach. The grid search method is used to obtain all combinations of the design variables in search space, then determine the objective functions for each suggested design variables s . The Pareto optimal approach is used to solve the conflict in the objective function of the design optimization problem.

3.2.1. Grid search method

The grid search method is a manual direct optimization method that involves forming a suitable grid using the design variables space S , where each grid point represents a suggested design variables $s = [p_1, p_2, \dots, p_m]$ [33].

The grid search is formulated by selecting upper bound and lower bound for each design variable as $[p_{1\max}, p_{1\min}], [p_{2\max}, p_{2\min}], \dots, [p_{m\max}, p_{m\min}]$ with step size of $\Delta p_1, \Delta p_2, \dots, \Delta p_m$, this allows to generate n equally spaced points for each design variable interval. Thus, the number of possible grid points n^m , where each point represents a possible solution of the design variables. Finally, the objective functions are calculated for each suggested design variables.

3.2.2. Pareto optimal approach

Pareto optimal approach is used to solve the multi-objective design optimization problem in machine design, for example, see [14,34]. The Pareto optimality is the condition where the one objective function cannot be minimized without maximizing the other objective function [35]. Consider ideal point of the Pareto optimal approach is given as is known as $\min f_1(s), \min f_2(s), \min f_3(s), \dots, \min f_Q(s)$, where Q is the number of objective functions. Then, the optimal solution can be obtained using the Euclidean distance as [36]

$$d = \sqrt{\sum_{q=1}^Q (f_q(s) - \min f_q(s))^2} \quad (14)$$

Fig. 4 shows the representation of the grid search and the Pareto optimal approach of optimization design problem with two design variables $s = [p_1, p_2]$ and two objective functions $f_1(s)$ and $f_2(s)$. First, the grid is generated, which consists of 16 points. Each variable has $n = 4$ equally points. Then, the two objective functions $f_1(s)$ and $f_2(s)$ are evaluated at each grid point. Finally, the optimal solution is determined by finding the shortest Euclidean distance between the Pareto optimal solution and the ideal point ($\min f_1(s), \min f_2(s)$), where the Euclidean distance for two objective functions is expressed as $d_2 = \sqrt{(f_1(s) - \min f_1(s))^2 + (f_2(s) - \min f_2(s))^2}$.

3.2.3. Conflict in the objective functions of reluctance actuator

The objective function of the design optimization of the reluctance actuator aims to maximize the actuator force and minimize the actuator mass to satisfy the requirements of the high precision motion system in the lithography machine. However, the actuator force depends on the cross-section area of the actuator and the number of turns in the coil. Thus, increasing the force requires increasing the actuator mass, so the objectives conflict in the design optimization problem can be presented by maximizing the force and minimizing the actuator mass. To solve the objectives conflict, the Pareto optimal approach is used. Consider the ideal point of the Pareto optimal approach is defined as $(\max(f_{k_E}), \min(f_M), \min(f_L), \min(f_R))$. Then, the Euclidean distances are calculated between the objective functions for the suggested design variables s and the ideal point as

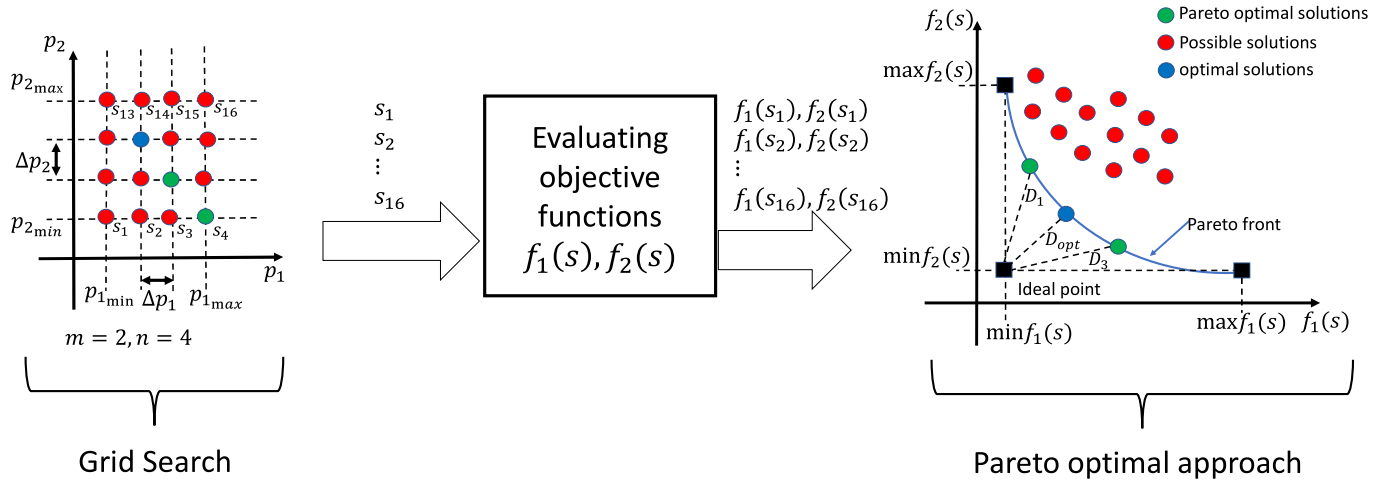


Fig. 4. Grid search and Pareto optimal approach representation of design optimization problem with two design variables $s = [p_1, p_2]$ and two objective functions $(f_1(s), f_2(s))$.

$$D(s) = \sqrt{\left(\frac{E_{k_E}(s)}{\min(E_{k_E})}\right)^2 + \left(\frac{E_M(s)}{\max(E_M)}\right)^2 + \left(\frac{E_L(s)}{\max(E_L)}\right)^2 + \left(\frac{E_R(s)}{\max(E_R)}\right)^2} \quad (15)$$

where $E_{k_E}(s) = f_{k_E}(s) - \max(f_{k_E})$, $E_M(s) = f_M(s) - \min(f_M)$, $E_L(s) = f_L(s) - \min(f_L)$, and $E_R(s) = f_R(s) - \min(f_R)$. The optimal design is selected with the shortest Euclidean distance as $D_{opt} = \min(D(s))$.

3.2.4. Design optimization steps

The optimal solution of the design variables is selected based on the shortest Euclidean distance of the Pareto optimal approach. The proposed optimization algorithm is illustrated by the flow chart in Fig. 5, and summarized in the following steps:

- Step 1:** Define the design requirements, including the desired force F_d , the air gap range $[g_{max}, g_{min}]$, and the coil American Wire Gauge (AWG#). The coil gauge refers to the coil diameter d and the maximum allowable current i_{mc} .
- Step 2:** Define the grid search space based on the range of the design variables. Consider the search space for the number of turns in the coil N is $[N_0, N_n]$ with a step size of ΔN and the search space of the width ratio r_b is $[r_{b0}, r_{bm}]$ with a step size of Δr_b .
- Step 3:** Evaluate the objective functions. Thereafter, the grid search of the design space is formed. Each grid point represents a suggested design variables $s = \{N, r_b\}$. The objective functions $f_{k_E}(s)$, $f_M(s)$, $f_L(s)$, and $f_R(s)$ are evaluated using the analytical expressions in (10), (11), (12), and (13), respectively.
- Step 4:** Select the elite designs. In this step, the magnetic force in (7) is calculated for each suggested design s , then only the designs $s_j = \{N, r_b\}$, where $j = 1, 2, 3, \dots, z$ that satisfy the constraint of $F_d \leq F \leq 1.01F_d$ are selected and they are called the elite designs. The objective functions for selected designs s_j are represented as a vector of $f_{k_E}(s) = \{f_{k_E}(s_1), \dots, f_{k_E}(s_z)\}$, $f_M(s) = \{f_M(s_1), \dots, f_M(s_z)\}$, $f_L(s) = \{f_L(s_1), \dots, f_L(s_z)\}$, and $f_R(s) = \{f_R(s_1), \dots, f_R(s_z)\}$.
- Step 5:** Select the optimal design. In this step, the designer may select the optimal design considering a slight advantage of the specific objective function over the others or using the Pareto optimal approach.

4. Optimal design example

This section presents a numerical design example is to verify the effectiveness of the proposed design optimization algorithm. Two approaches are considered: First, the optimal design for specific different coil wire gauges (AWG#), thus, the optimal parameters are $s = \{N, r_b\}$. Second, the coil diameter d is included in the optimization algorithm, hence, the optimal design variables become $s = \{N, r_b, d\}$.

4.1. Optimal design for specific coil wire gauge

The optimal design for a specific coil wire gauge (AWG#22) with $i_{mc} = 7A$ and $d = 0.6542mm$ is obtained through the following design steps

- Step 1:** The design requirements are: (i) the desired force of $F_d = 400N$, (ii) the coil wire gauge is AWG#22 with $d = 0.6452mm$ and $i_{mc} = 7A$, and (iii) the range of the air gap is $g_{max} = 2.1mm$ and $g_{min} = 0.1mm$. The air gap range is selected to satisfy a short stroke displacement of 2 mm, and to avoid the direct contact between the mover and the stator of the reluctance actuator.
- Step 2:** The search spaces are defined to generate the grid search space. The search space of N is $N_0 = 100$ and $N_n = 1000$ with $\Delta N = 1$, and the search space of r_b is $r_{b0} = 0.01$, $r_{bm} = 3$ with step size of $\Delta r_b = 0.01$. These search spaces are selected to avoid the large increase in the actuator mass and to ensure that the reluctance actuator will generate the desired force.
- Step 3:** The objective functions $f_{k_E}(s)$, $f_M(s)$, $f_L(s)$, and $f_R(s)$ are evaluated for each suggested design variables $s = \{N, r_b\}$ (grid points) considering the maximum air gap $g_{max} = 2.1mm$ with the input current of $i_{mc} = 7$. Fig. 6 shows the contour plot of the objective functions in terms of the design variables $s = \{N, r_b\}$, where each grid point represents a suggested design for the reluctance actuator. These plots can be used as design charts to determine the design variables of the reluctance actuator based on the design requirements.
- Step 4:** Select the elite designs. The magnetic force in (7) is calculated for the suggested design variables s and Fig. 7 illustrates contour plot of the magnetic force term of the design variables s . The selected designs for the reluctance actuator that satisfy the desired force condition of $F_d \leq F \leq 1.01F_d$ are called the elite designs and presented by the red points in Fig. 7.
- Step 5:** Select the optimal design from the elite designs obtained in Step 4 using the Pareto optimal approach.

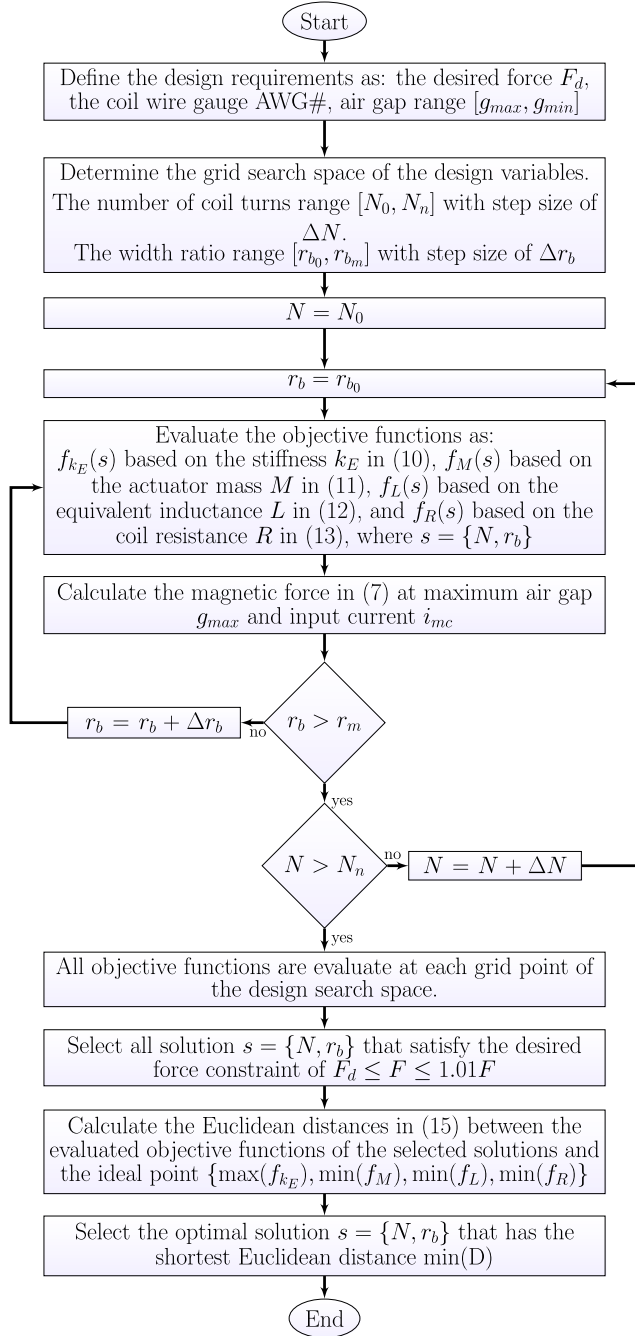


Fig. 5. Flow chart of the proposed design optimization algorithm to obtain the optimal design of the reluctance actuator.

Using the Pareto optimal approach, the optimal design is obtained by finding the minimum Euclidean distance D_{opt} between all elite designs and the ideal point using (15). The ideal point in the Pareto optimal approach presents the best values of the objective functions as $\max(f_{k_E}) = -0.3797 \text{ N}/\mu\text{m}$, $\min(f_M) = 0.6531 \text{ kg}$, $\min(f_L) = 34.36 \text{ mH}$, and $\min(f_R) = 3.384 \Omega$. Then, by calculating the Euclidean distance it can be found that the optimal design of the reluctance actuator is obtained with the design variables of $N = 574$ and $r_b = 1.27$. The objective functions of the optimal design variables are $f_{k_E} = -0.3801 \text{ N}/\mu\text{m}$, $f_M = 0.655 \text{ kg}$, $f_L = 34.49 \text{ mH}$, and $f_R = 3.38 \Omega$. Fig. 8 shows the Pareto optimal approach representation for the stiffness f_{k_E} and the actuator mass f_M objective functions. In this figure, the elite

Table 2

The optimal design variables and the objective functions for different coil wire gauges of AWG#22, AWG#21, and AWG#20.

	AWG#23	AWG#22	AWG#21	AWG#20
N	796	574	416	319
r_b	1.3	1.27	1.62	2.08
$f_{k_E} (\text{N}/\mu\text{m})$	-0.380	-0.380	-0.380	-0.380
$f_M (\text{kg})$	0.7868	0.6553	0.6796	0.7422
$f_L (\text{mH})$	76.3	34.4	20.8	13.92
$f_R (\Omega)$	6.39	3.38	2.13	1.44
$i_{mc} (\text{A})$	4.7	7	9	11

Table 3

The optimal design of the reluctance actuator for coil wire gauge of AWG#20.

Design variables	value	Objective function	value
N	314	$f_{k_E} (\text{N}/\mu\text{m})$	-0.375
r_b	2.34	$f_M (\text{kg})$	0.7778
$d (\text{mm})$	0.8128	$f_L (\text{mH})$	14.1
$a (\text{mm})$	15	$f_R (\Omega)$	1.5
$b (\text{mm})$	35.1		
$w_c (\text{mm})$	14		
$h_c (\text{mm})$	26		
$H (\text{mm})$	42		
$W (\text{mm})$	95.1		

designs are the red points, the Pareto optimal solutions are the green points, the ideal point is the black point, and the optimal design is the blue point. As shown in Fig. 8, the optimal design (blue point) is the nearest point to the ideal point (black point).

Using the same design steps, the optimal design variables of the reluctance actuator are obtained for different coil wire gauges of AWG#23 ($i_{mc} = 4.7 \text{ A}$, $d = 0.574 \text{ mm}$), AWG#22 ($i_{mc} = 9 \text{ A}$, $d = 0.7239 \text{ mm}$), and AWG#20 ($i_{mc} = 11 \text{ A}$, $d = 0.8128 \text{ mm}$). The design variables and the objective functions are presented in Table 2. The table shows that the objective functions f_M , f_L , and f_R are changed with the coil wire gauge. In the next section, the grid search is modified to consider the coil diameter d to be one of the optimized design variables.

4.2. Optimal design considering the coil wire diameter within the design variables

In this section, the optimization algorithm is modified to determine the coil diameter d within the design variables. The design variables to be optimized are the number of turns N , the width ratio r_b , and the coil diameter d , then each point in the grid presents a suggested design as $s = \{N, r_b, d\}$. Consider the search space of N is $N_0 = 100$ and $N_n = 1000$ with $\Delta N = 1$, the search space of r_b is $r_{b0} = 0.01$, $r_{bm} = 3$ with a step size of $\Delta r_b = 0.01$, and the coil diameters are $d = [0.4547, 0.5105, 0.5740, 0.6452, 0.7239, 0.8128]$. The maximum allowable currents for each coil diameter are $i_{mc} = [2.7, 3.5, 4.7, 7, 9, 11]$. Following the design steps, the objective functions are calculated for all suggested designs variables $s = \{N, r_b, d\}$; then, the Pareto optimal approach is used to determine the optimal design. Fig. 9 shows the Pareto optimal approach representation using different objective functions. In this figure, the black point presents the ideal point in the Pareto optimal approach which has the best values of the objective function as $\max(f_{k_E}) = -0.384 \text{ N}/\mu\text{m}$, $\min(f_M) = 0.648 \text{ kg}$, $\min(f_L) = 14.03 \text{ mH}$, and $\min(f_R) = 1.371 \Omega$. The red points present the elite designs that satisfy the force condition $F_d \leq F \leq 1.01F_d$. The optimal design presented by the blue point is determined by finding the minimum Euclidean distance (15) between the elite designs (red points) and the ideal point (black point). The design variables of the optimal design are the number of turns of $N = 314$, the width ratio of $r_b = 2.34$ and the coil diameter of $d = 0.8182 \text{ mm}$ (AWG#20). The design variables and the objective functions of the optimal design are given in Table 3.

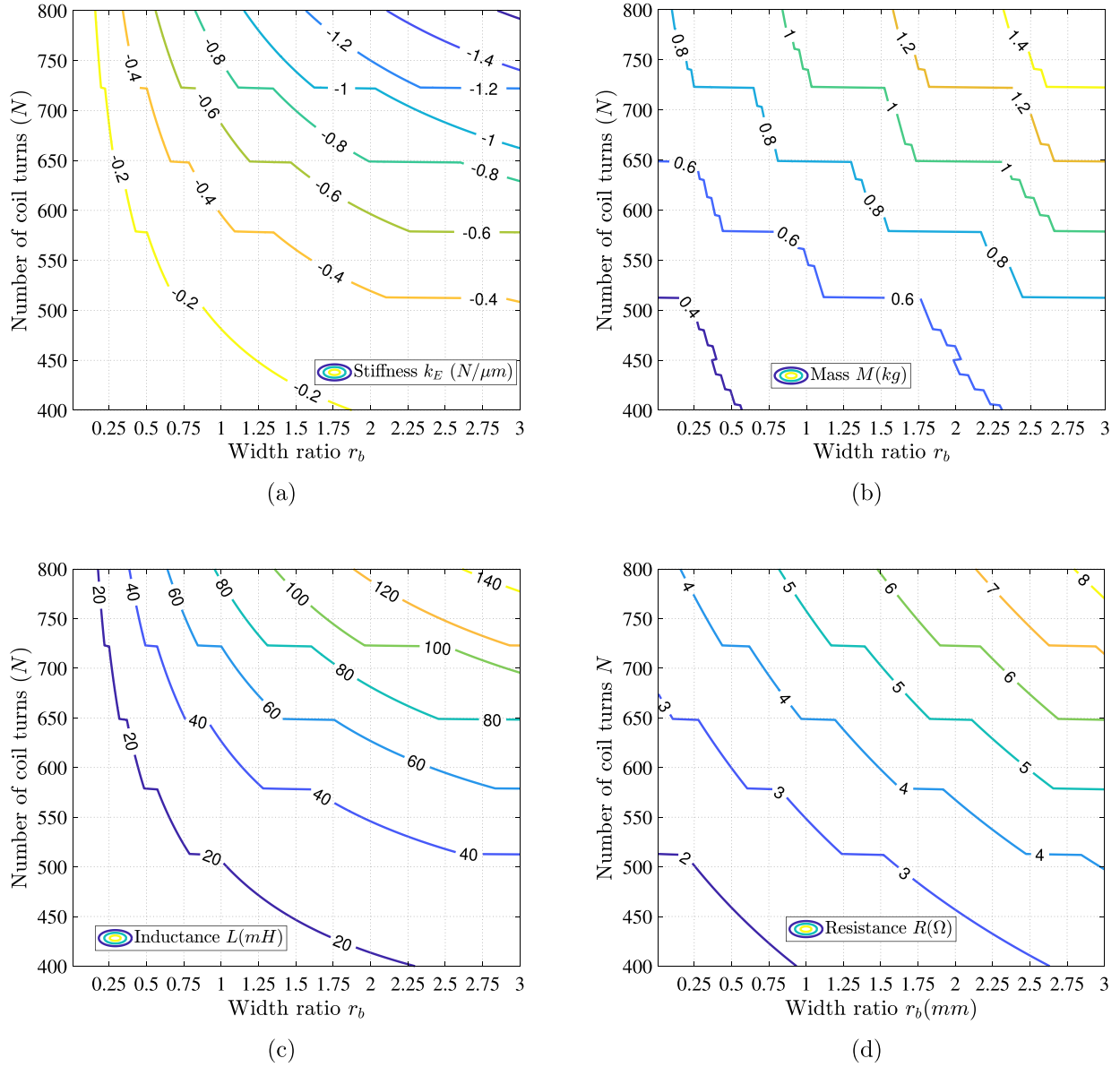


Fig. 6. The contour plot representation of the objective functions in terms of the design variables $s = \{N, r_b\}$ for (a) the stiffness f_{k_E} (N/μm) based on (10), (b) the actuator mass f_M (kg) based on (11), (c) the equivalent inductance f_L (mH) based on (12), and (d) The coil resistance f_R (Ω) based on (13).

4.2.1. Design with different search space

In this study, the focus is to formulate the design variables of the reluctance actuator as a multi-objective design optimization problem. Different multi-objective optimization algorithms with different search methods can be used to obtain the optimal design of the reluctance actuator. In this study, the design problem is three dimensions optimization problem. Then, all combinations of the design variables within the specific search space can be obtained using grid search optimization. The effectiveness of the grid search is measured by assuming different search spaces and obtaining the optimal design variables. Table 4 shows the comparison between the optimal design variables and the correspond objective functions for the different search space. The results show that the different optimal designs with different actuator masses are obtained.

4.2.2. Performance metrics for the proposed optimization algorithm

The performance metrics are indicators to measure the performances of the multi-objective optimization algorithm, including convergence, coverage, and success [37]. Different performance indicators

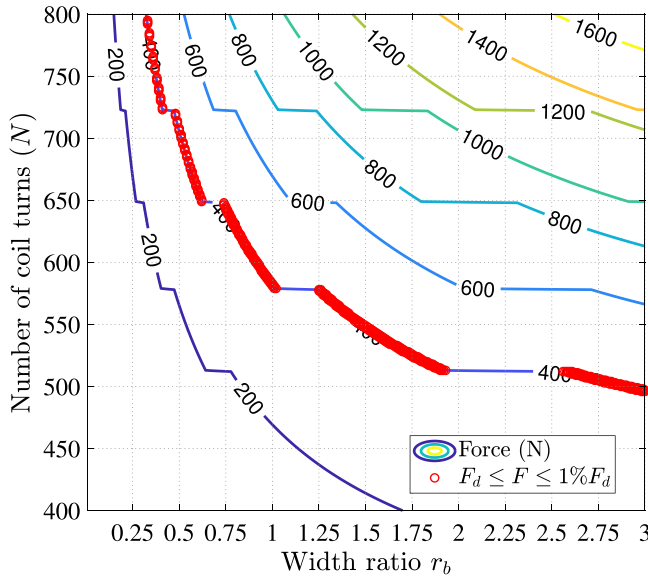
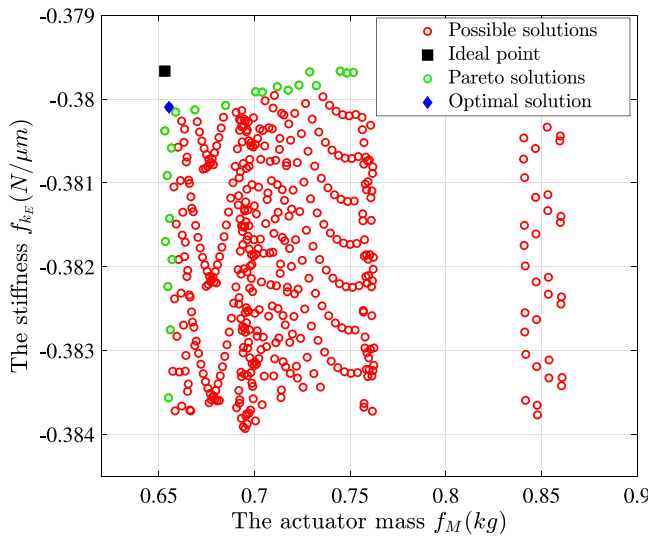
can be used to measure the algorithm performances, for example, Generation Distance (GD) and Hypervolume for convergence performance, Diversity Measure and Maximum Spread for coverage performance, and Error Ratio (ER) for success performance [37]. In this study the performances indicators are

1. Generation Distance DG: This indicator calculates the Euclidean distance D_i between the i th obtained Pareto optimal solution and the nearest solutions on the Pareto front (Reference points) [38]. Mathematically, the generation distance is formulated as $GD = \frac{\sqrt{\sum_{i=1}^q D_i^2}}{q}$, where q is the number of obtained Pareto optimal solution. For the proposed grid method, the Euclidean distance is given by (15) and the reference points are selected as shown in Fig. 9(a), then the GD is calculated between all obtained solutions and the reference and is equal to $GD = 0.156$.
2. Maximum Spread: this indicator calculate the maximum diagonal distance and present a hypercube using the Pareto optimal front [39]. Mathematically the maximum spread can be

Table 4

The optimal design variables and the objective functions of the reluctance actuator considering different search space.

Grid #	Search space	N	rb	d	f_{k_E}	f_M	f_L	f_R
Grid #1	$N_0 = 100, N_n = 1000, \Delta N = 1$ $r_{b0} = 0.01, r_{bn} = 3, \Delta r_b = 0.01$	314	2.34	0.8182	-0.375	0.7778	14.1	1.50
Grid #2	$N_0 = 100, N_n = 1000, \Delta N = 5$ $r_{b0} = 0.01, r_{bn} = 3, \Delta r_b = 0.01$	320	2.16	0.8182	-0.375	0.7549	14.1	1.47
Grid #3	$N_0 = 100, N_n = 1000, \Delta N = 10$ $r_{b0} = 0.01, r_{bn} = 3, \Delta r_b = 0.01$	300	2.93	0.8182	-0.375	0.8461	14.1	1.61
Grid #4	$N_0 = 100, N_n = 1000, \Delta N = 1$ $r_{b0} = 0.01, r_{bn} = 3, \Delta r_b = 0.02$	323	2.07	0.8182	-0.375	0.7433	14.1	1.46
Grid #5	$N_0 = 100, N_n = 1000, \Delta N = 5$ $r_{b0} = 0.01, r_{bn} = 3, \Delta r_b = 0.02$	310	2.49	0.8182	-0.375	0.7891	14.1	1.53
Grid #6	$N_0 = 100, N_n = 1000, \Delta N = 10$ $r_{b0} = 0.01, r_{bn} = 3, \Delta r_b = 0.02$	320	2.17	0.8182	-0.375	0.7564	14.1	1.48

**Fig. 7.** The magnetic force calculated using (7) interm of the suggested desing variables (N and r_b), where the red points present the elite designs that satisfy the desired force condition $F_d \leq F \leq 1.01F_d$.**Fig. 8.** Pareto optimal approach representation of two objective functions, the stiffness f_{k_E} and the actuator mass f_M .

calculated by $M = \sqrt{\sum_{i=1}^4 \max(D(a_i, b_i))}$, where D is the euclidean distance, a_i and b_i are the maximum and the minimum values of the objective i , respectively. For the proposed grid method, the values of a, b as pair for each objective function are $f_{k_E} = \{-3.7476 \times 10^5, -3.8397 \times 10^5\}$, $f_M = \{2.092, 0.6480\}$, $f_L = \{139.055, 14.0257\}$, and $f_R = \{12.5432, 1.3710\}$. Then, the maximum spread is calculated using the Euclidean distance (15) between the maximum and minimum points as $M = \sqrt{0.0246 + 0.6902 + 0.8991 + 0.8907} = 1.5826$.

3. Error Ratio: This indicator measures the approximation of the obtained solution to the Pareto front reference set [40]. Mathematically, this indicator is calculated by count the number of Pareto optimal solutions belong the Pareto front (Reference points) and divides it to the total number of solution as $ER = \frac{\sum_{i=1}^n e_i}{n}$, where $e_i = 1$ if the obtained solution belong the Pareto front and $e_i = 0$ otherwise. For the proposed grid search, the total number of the obtained solutions is 2536, and the number of the solution closest to the reference points is 40, then the error ratio is $ER = \frac{40}{2536} = 0.00157$.

5. Optimal design verification

This section presents a verification for the optimal design using analytical model simulation, Finite Element Analysis (FEA) simulation, and simulation with the motion system. Also, this section presents the sensitivity analysis for the optimized design variables.

5.1. Analytical and FEA models simulation with fixed air gap

The simulation results are obtained for fixed air gaps of $g = 0.1$ mm, $g = 1$ mm, and $g = 2.1$ mm with the design variables given in Table 3. The simulation results are obtained using the dynamic model (8) and FEA model with the optimal design variables. The optimal design variables are selected to ensure that the reluctance actuator is operated at the linear region of the BH-curve. In this study, the ferromagnetic material of E-core and I-core of the reluctance actuator is Cobalt Steel (Vacoflux 50), which has the BH-curve given in Fig. 10(a) with the saturation flux density of $B_{sat} = 2.3$ T. The relative permeability of the ferromagnetic material is calculated for each segment of BH-curve as shown in Fig. 10(b). The reluctance actuator is designed to operate at linear region with the relative permeability about of $\mu_r(B, H) = \mu_r = 11664$. For the constant relative permeability, the required input current i_d to generate the desired force F_d at given air gap g can be calculated using (7) as

$$i_d = \left[\frac{\frac{1}{r_b} l_{s1} + \frac{1}{2} l_{s2} + \frac{1}{4} l_m}{\mu_r} + \left(\frac{1}{r_b} + \frac{1}{2} \right) g \right] \sqrt{\frac{2F_d}{(\mu_0 A_{aa} N^2) \left(\frac{1}{r_b} + \frac{1}{2} \right)}} \quad (16)$$

This equation is used to determine the desired current to generate the desired force of $F_d = 400$ N at different air gaps as shown in Table 5.

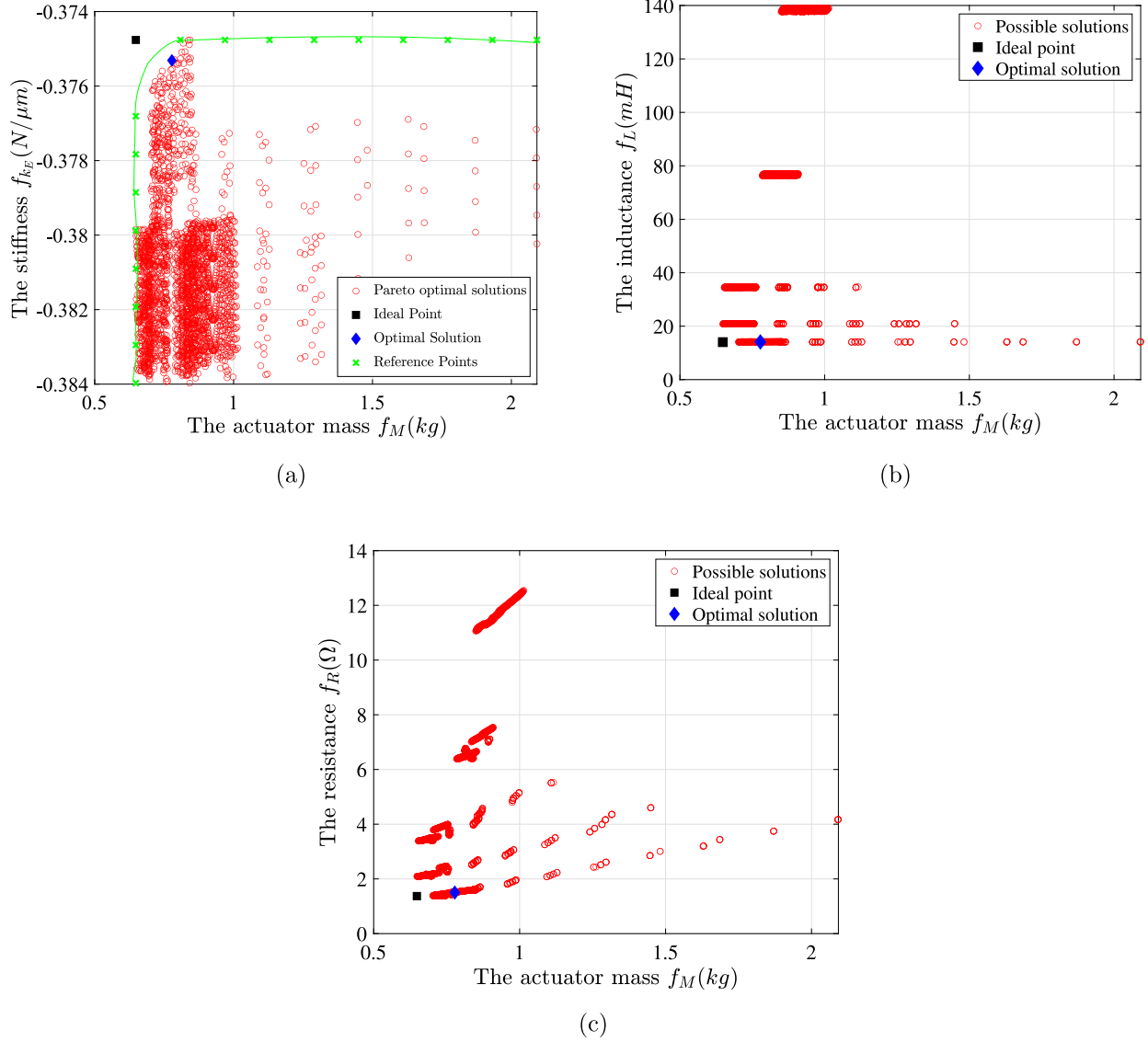


Fig. 9. Pareto optimal approach representation of two objective functions: (a) the actuator mass f_M and the stiffness f_{k_E} , (b) the actuator mass f_M and the inductance f_L , and (c) the actuator mass f_M and the resistance f_R .

Table 5

The characteristic of the optimal design of the reluctance actuator with the desired input current i_d at different air gaps.

Air gap g (mm)	Current i_d (A)	Actuator Stiffness k_E (μm/N)	Actuator Mass M (kg)	Equivalent Inductance L (mH)	Equivalent Resistance R (Ω)
0.1	0.55	-7.5	0.7778	281.8	1.5
1	5.19	-0.7947	0.7778	29.9	1.5
2.1	10.87	-0.3797	0.7778	14.3	1.5

For FEA simulation, the reluctance actuator in Fig. 1 with the optimal design variables in Table 3 is implemented using COMSOL software. The simulation results are obtained using magnetostatic model to study the magnetic force and the magnetic flux density at different air gaps of $g = \{0.1, 1, 2.1\}$ mm with input currents of $i = \{0.55, 5.19, 10.87\}$ A. The magnetic force is obtained as function of air gap under different input current as shown in Fig. 11(a). This figures illustrates a good matching between the magnetic force obtained by the analytical model (7) and the FEA model with error about 7% at maximum air gap as shown in Fig. 11(b). Fig. 12 shows the magnetic flux density $B(T)$ distribution of the optimal design of the reluctance

actuator. It can be observed that the magnetic flux $B(T)$ values are less than the saturation flux $B_{sat} = 2.3$ T. It can be noticed that the maximum obtained magnetic flux is about $B = 1.55$ T at air gap $g = 2.1$ mm, then the reluctance actuator has a relative permeability of the linear region of BH-curve. Thus, selecting the optimal design variables can be used to linearize the dynamic behavior of the reluctance actuator.

The dynamic responses of the reluctance actuator (current and force) are obtained by solving the dynamic model (8) and using the magnetodynamic FEA model at a fixed air gap g . The applied voltage is a DC voltage with amplitudes of $v = 0.825$ V, $v = 7.785$ V, and $v = 16.31$ V. These voltages able to generate a steady-state current with

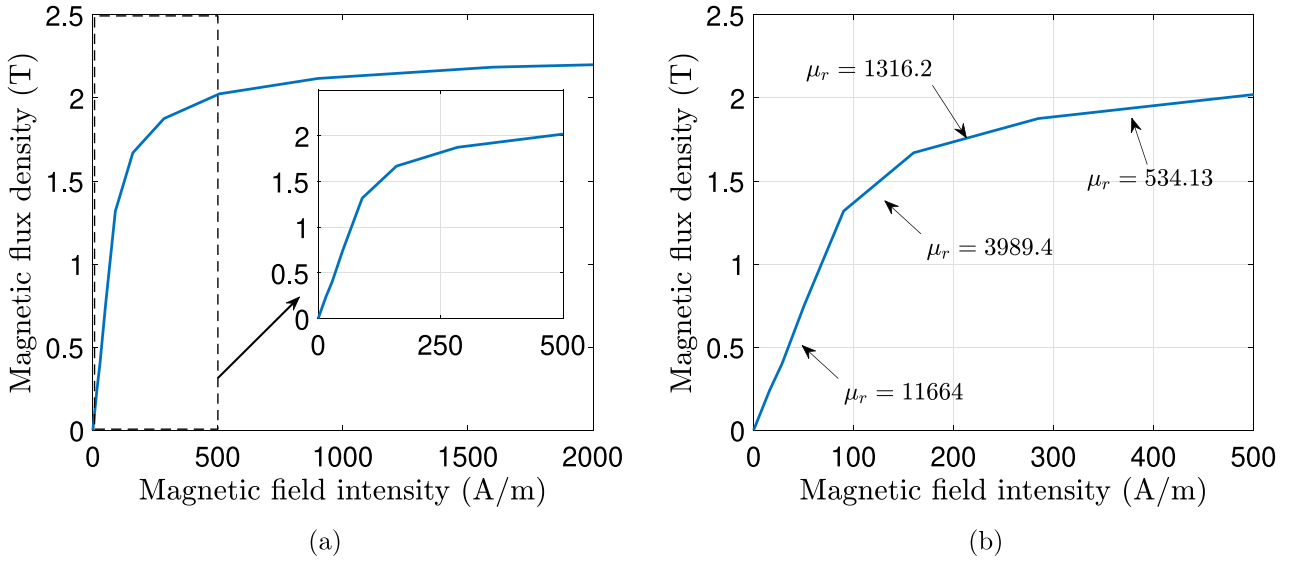


Fig. 10. (a) The BH curve of the ferromagnetic material Cobalt Steel Vacoflux 50, and (b) linear region of BH curve with corresponding relative permeability μ_r .

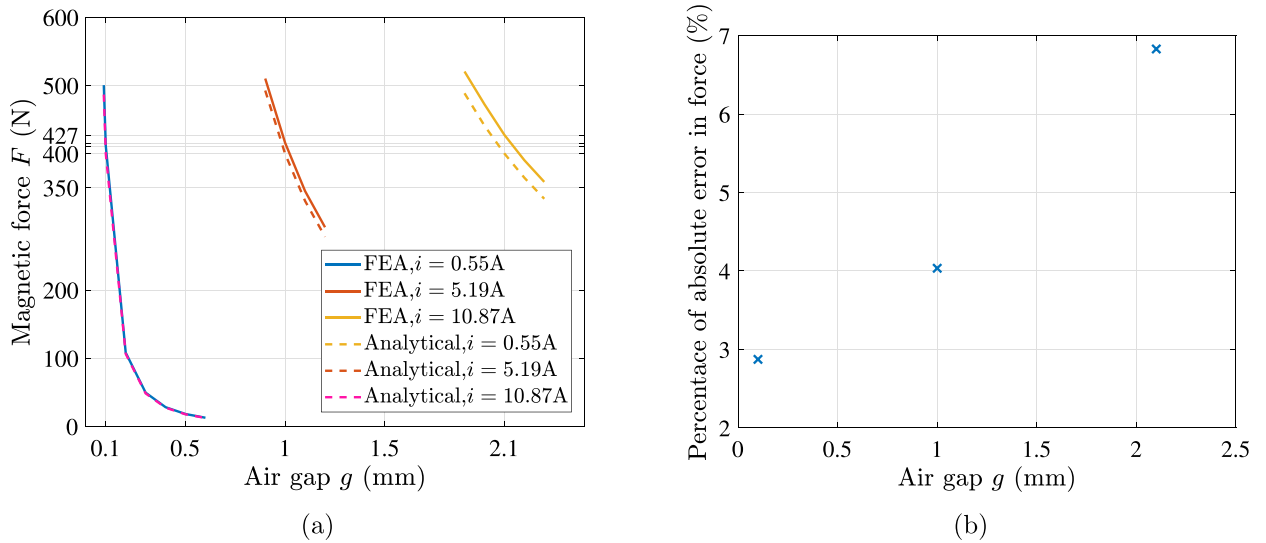


Fig. 11. (a) The force-to-air-gap relationship using COMSOL FEA expression in (7) with magnetic permeability of $\mu_r = 11664$, and (b) the percentage of absolute error in the magnetic force using two methods.

values of $i_{ss} = 0.55$ V, $i_{ss} = 5.19$ A, and $i_{ss} = 10.87$ A at air gaps of $g = 0.1$ mm, $g = 1$ mm, and $g = 2.1$ mm, respectively. Fig. 13 and Fig. 14 show the time-domain responses of the input current and the magnetic force of the reluctance actuator at different air gaps, respectively. From the figures, it can be noticed that the reluctance actuator with the optimal design variables can generate the desired force at different air gaps and input currents. Also, it can be noticed that the time constant of the dynamic response increases by decreasing the air gap. By comparing the time constant of the current response is faster than the time constant of the force response. Fig. 15 shows the magnetic flux distribution using the FEA magnetodynamic model. This figure shows that the magnetic flux values at different air gaps are less than the saturation value.

5.2. Analytical model simulation with motion system

The simulation results in the previous section verify that the optimal design of the reluctance generates the desired force, and it is operated at the linear region of the BH-curve. In this section, the time-domain response of the reluctance actuator motion system attached to the motion system is obtained using the dynamic model (8). The design variables of the reluctance actuator is given in Table 3 and the motion system has displacement range of $x = [0, 2]$ mm and force of $F = 400$ N. The motion system is modeled as a spring-mass-damper system with a spring coefficient of $k = F/x = 400/2 = 200$ N/mm. The mass of the motion system included the mover of the reluctance actuator is about $m = 505$ g, then the system natural frequency is about $f_n = \frac{1}{2\pi} \sqrt{\frac{k}{m}} \approx 100$ Hz. To makes the motion system with a damping ratio of $\zeta = 0.707$, the damping coefficient is selected as $b = 450$ Ns/m.

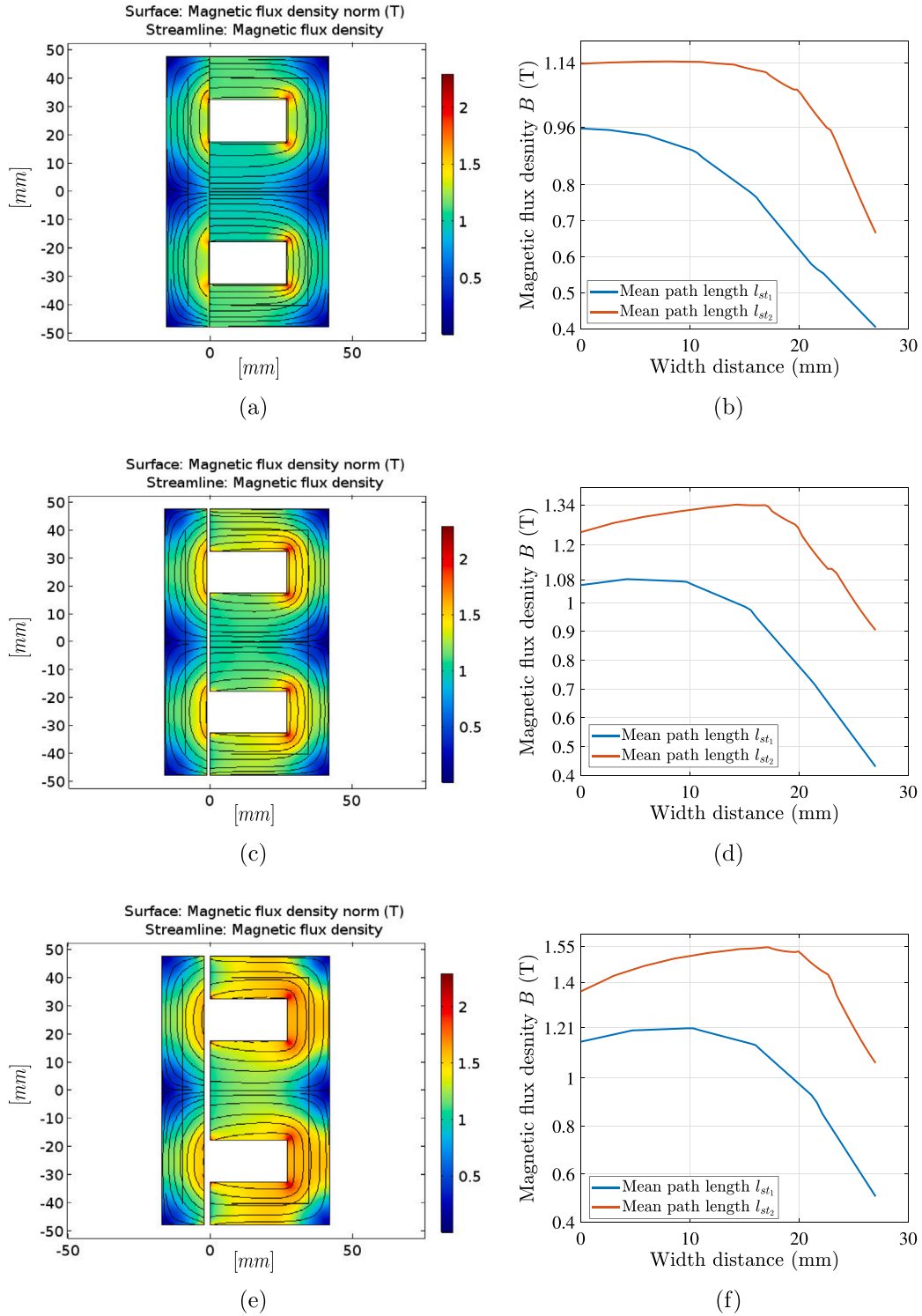


Fig. 12. (a) The BH curve of the ferromagnetic material Cobalt Steel Vacoflux 50. Magnetic flux density $B(T)$ of the optimal design of the reluctance actuator with parameters in Table 3 using COMSOL for: (b) $g_{min} = 0.1$ mm with $i = 0.55$ A, (c) $g = 1$ mm with $i = 5.19$ A, and (d) $g_{max} = 2.1$ mm with $i = 10.87$ A.

In this simulation, the applied voltages are selected where the magnetic force is limited by the mechanical force. Fig. 16 shows the responses of the reluctance actuator motion system (current, force, and displacement) under step input voltage with amplitude of $V = 9$ V, 10 V, and 11 V. For these inputs values, the reluctance actuator generates forces about 48 N, 65 N, 92 N, which allows moving the stage with output displacements about 0.24 mm, 0.32 mm, 0.46 mm, respectively.

It can be noticed that the air gaps are about 1.86 mm, 1.78 mm, and 1.64 mm.

Also, the responses of the reluctance actuator motion system (current, force, and displacement) are obtained under harmonic input voltage as $v(t) = V_p \sin(2\pi ft)$ with peak value of $V_p = 9$ V, 10 V and 11 V. Fig. 17 shows the time-domain result for input voltage under excitation frequency of $f = 1$ Hz and $f = 10$ Hz in Fig. 17 and Fig. 18 shows the time-domain result for input voltage under excitation frequency of

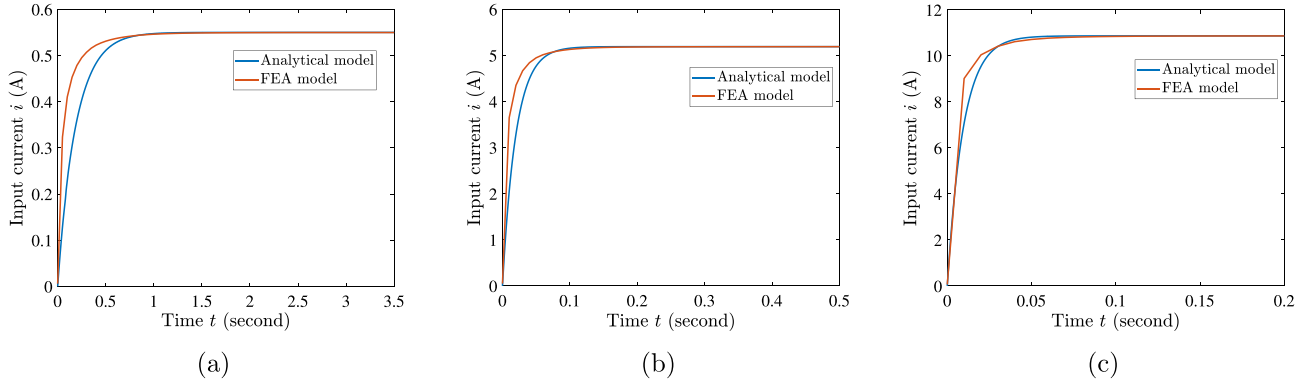


Fig. 13. Time-domain response of the input current of the reluctance actuator using analytical model and using magnetodynamic model in FEA at (a) air gap $g = 0.1$ mm, (b) air gap $g = 1$ mm, and (c) air gap $g = 2.1$ mm.

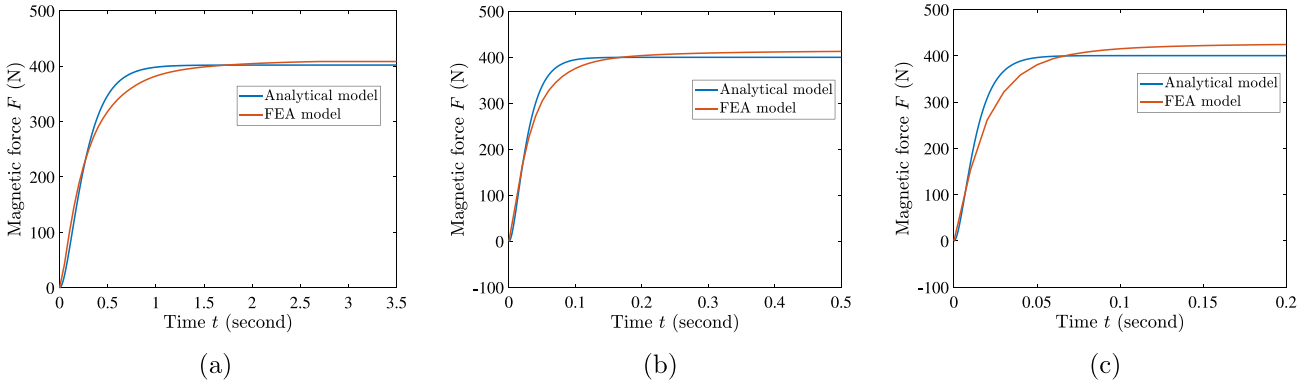


Fig. 14. Time-domain response of the magnetic force of the reluctance actuator using analytical model and using magnetodynamic model in FEA at (a) air gap $g = 0.1$ mm, (b) air gap $g = 1$ mm, and (c) air gap $g = 2.1$ mm.

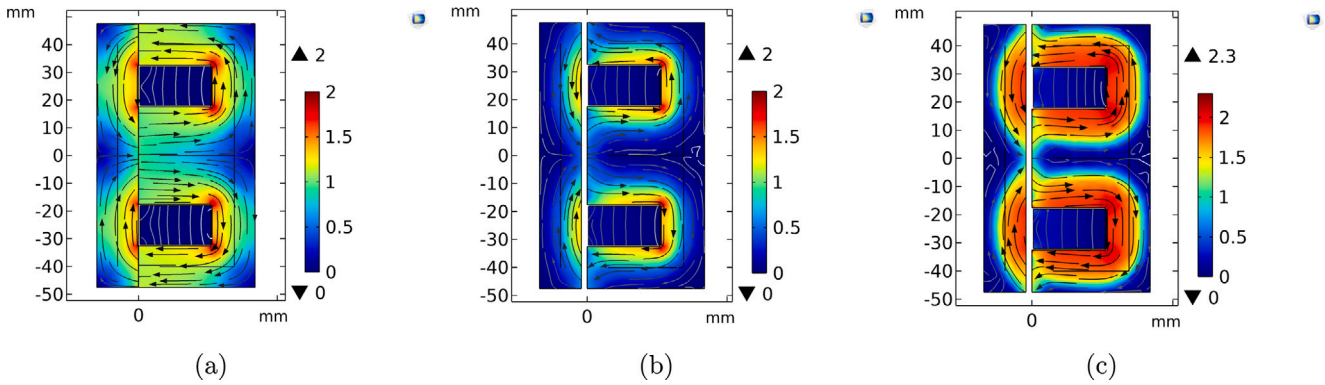


Fig. 15. Magnetic flux distribution of the reluctance actuator using magnetodynamic model in FEA at (a) air gap $g = 0.1$ mm at $t = 3.5$ s, (b) air gap $g = 1$ mm at $t = 0.5$ s, and (c) air gap $g = 2.1$ mm at $t = 0.2$ s.

$f = 10$ Hz. The two figures indicate that the output force F in the air gap depends on the magnitude V_p and the frequency f of the input voltage $v(t)$.

5.3. Sensitivity analysis of the optimal design variables

In optimization problems, the sensitivity analysis studies the change of the objective functions of the optimal design to the change in the optimized design variables parameters [41]. In this section, we study the effects of having slight variations of $\pm 1\%$ in the designed variables (N , and r_b) on the values of the objective functions. The objective functions and the desired currents with slight variation of $\pm 1\%$ of the optimal design variables of the reluctance actuator are calculated as

shown in Table 6. According to these results, increasing the number of turns N by almost 1% (from $N = 314$ to $N = 318$) decrease the desired current i_d by almost 1%, and increase the objective function f_M by almost 0.35%, increase the objective function f_L by almost 2%, and increase the objective function f_R by almost 1%. Decreasing the number of turns N by 1% (from $N = 314$ to $N = 311$) increases the desired current i_d by almost 1%, and decrease the objective function f_M by almost 0.35%, decrease the objective function f_L by almost 2.1%, and decrease the objective function f_R by almost 1.3%.

Also, increasing the width ratio r_b by almost 2% (from $r_b = 2.34$ to $r_b = 2.38$) decrease the desired current i_d by almost 0.35%, and increase the objective function f_M by almost 1%, increase the objective function f_L by almost 1%, and increase the objective function f_R by almost

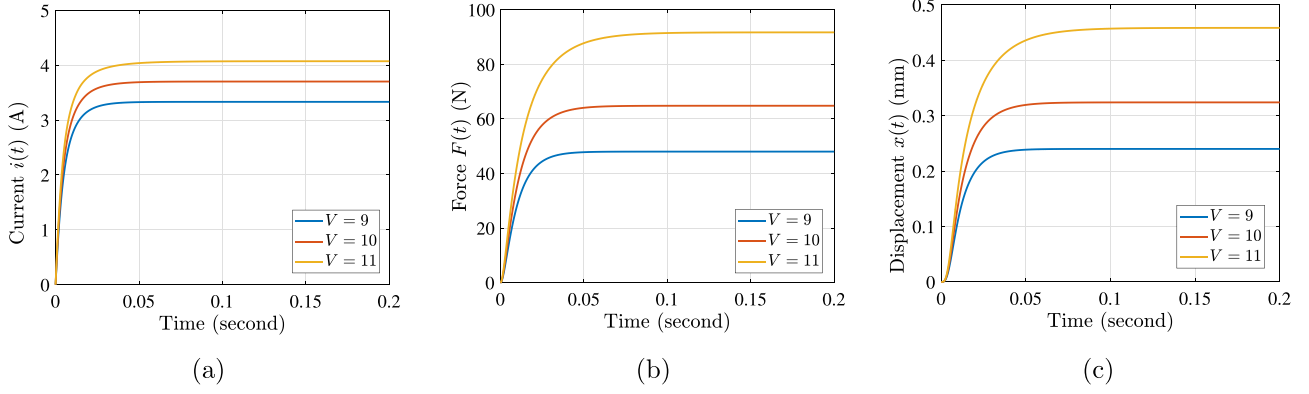


Fig. 16. The output force and displacement of the reluctance actuator motion system under input $v(t)$: (a) the input current in the coil $i(t)$, (b) the generated force in the air gap $F(t)$, and (c) the displacement of the movable stage $x(t)$.

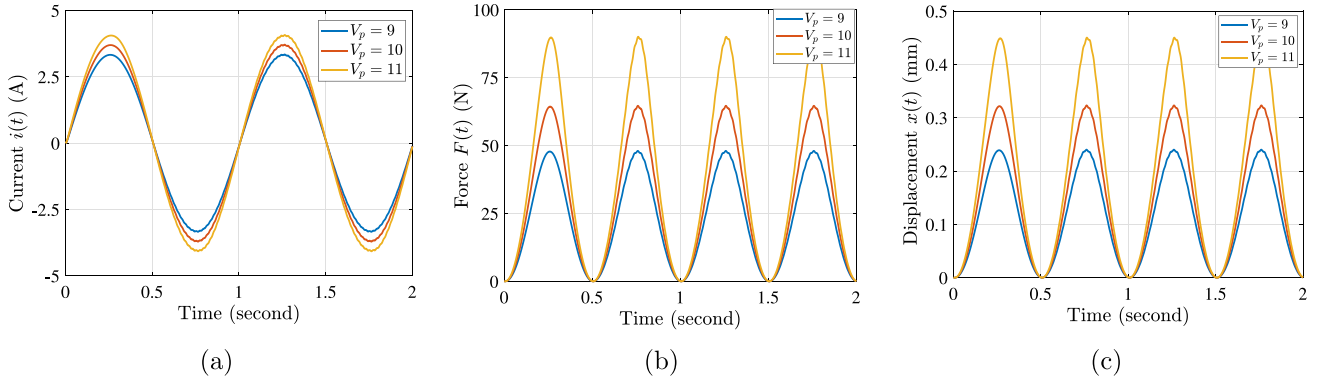


Fig. 17. The output force and displacement under the input voltage of $v(t) = V_p \sin(2\pi ft)$, where $f = 1 \text{ Hz}$: (a) the input current in the coil $i(t)$ (A), (b) the generated force in the air gap $F(t)$ (N), and (c) the displacement of the movable stage $x(t)$.

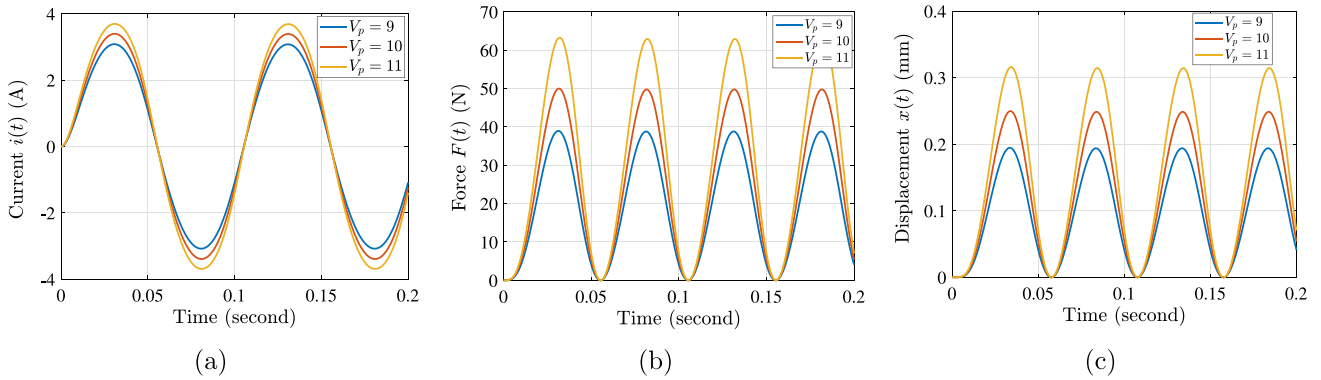


Fig. 18. The output force and displacement under the input $v(t) = V_p \sin(2\pi ft)$, where $f = 10 \text{ Hz}$: (a) the input current in the coil $i(t)$, (b) the generated force in the air gap $F(t)$, and (c) the displacement of the movable stage $x(t)$.

Table 6

The objective functions and the desired current with slight variation in the design variables of the reluctance actuator.

Number of turns N	Width ratio r_b	Current i_d (A)	Objective f_{k_E} ($\mu\text{m}/\text{N}$)	Objective f_M (kg)	Objective f_L (mH)	Objective f_R (Ω)
314	2.34	10.87	-0.3797	0.7778	14.3	1.50
311	2.34	10.97	-0.3798	0.7678	14.0	1.49
318	2.34	10.73	-0.3797	0.7805	14.6	1.52
314	2.38	10.83	-0.3797	0.7838	14.4	1.51
314	2.29	10.92	-0.3797	0.7704	14.1	1.49

0.65%. Decreasing the width ratio r_b by almost 2% (from $r_b = 2.34$ to $r_b = 2.29$) increases the desired current i_d by almost 1%, and decrease the objective function f_M by almost 1%, decrease the objective function

f_L by almost 1%, and decrease the objective function f_R by almost 0.65%.

6. Conclusion

This study proposed an optimization algorithm based on the grid search method and Pareto optimal approach to select the design variables, including the number of coil turns and the width ratio of the actuator poles of the E-core reluctance actuator. First, the characteristics of the reluctance actuator were expressed in terms of the actuator design variables. After that, the objectives of the optimization algorithm were formulated to drive the actuator stiffness value to zero, minimize the actuator mass, minimize the inductance, minimize the coil resistance. The optimization algorithm was used to determine the optimal design variables of the reluctance actuator based on the desired specifications. This design was obtained for a specific coil wire gauge, and then it was obtained considering the diameter of the coil within the optimization algorithm. The obtained optimal design was implemented in a precision motion system to drive the movable stage. The magnetic flux distribution of the optimal design of the reluctance actuator was obtained using FEA in COMSOL. The results showed that the reluctance actuator could generate the desired force at different air gaps where the magnetic flux value is less than the saturation value. The dynamic behavior of the precision motion system was verified using lumped modeling in Simscape. The results in the time domain showed that the stable displacement and force responses for different applied voltages. Future work will consider a control system to enhance the dynamic response specification and track the desired force and displacement.

CRediT authorship contribution statement

Mohammad Al Saaideh: Conception and design of study, Acquisition of data, Analysis and/or interpretation of data, Writing – original draft. **Natheer Alatawneh:** Conception and design of study, Acquisition of data, Analysis and/or interpretation of data, Writing – original draft. **Mohammad Al Janaideh:** Conception and design of study, Acquisition of data, Analysis and/or interpretation of data, Writing – original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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